
Synthetic Cultural Agents 2.0: Beyond Persona-Prompting

A Position Paper on Structural Preference Estimation for Culturally Aligned Language Models

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Status. Intermediate summary of SCA 2.0: structural conjecture, implemented DPO pipeline, Tier-1 synthetic pair recovery, and Tier-2 out-of-sample empirical evaluation on World Values Survey Wave 7 items ($N = 35$). Code & data: github.com/EconLLM-Lab/SCA2_PofW.

Abstract

Large language models are increasingly used as simulators of human judgment in economics and adjacent social sciences, yet their default alignment reflects predominantly Western, Educated, Industrialized, Rich, and Democratic (WEIRD) preference data. Earlier Synthetic Cultural Agents (SCA 1.0) addressed this gap by conditioning frontier models on ethnographic profiles at inference time. That approach is useful for piloting but is reduced-form: cultural signal lives only in the prompt, is fragile to model updates, and does not support formal identification or counterfactual analysis.

This position paper outlines SCA 2.0, a structural alternative. We treat Direct Preference Optimization (DPO) as maximum-likelihood estimation in a Bradley–Terry random-utility model, using the Global Preferences Survey (GPS) six-dimensional cultural state vector to generate culture-conditioned synthetic preference data. A student model (Llama-3.1-8B-Instruct) is fine-tuned via QLoRA with unconditioned prompts (`PROMPT_COUNTRY_CONDITIONING = False`) so that cultural dispositions are encoded into parameters rather than transient context.

We evaluate the framework across two tiers: (i) Tier-1 synthetic pair recovery ($n = 200$), where own-country adapters achieve 70.0–78.5% accuracy; and (ii) Tier-2 empirical out-of-sample evaluation on $N = 35$ unseen World Values Survey (WVS) Wave 7 items mapped to GPS dimensions. On non-WEIRD targets (Mexico), the matched MEX adapter demonstrates strong cultural specificity, outperforming the cross USA adapter on 97.1% of questions ($\Delta\text{TVD} = 0.1495$, 95% CI: $[0.0964, 0.2152]$). Conversely, on US targets, the USA adapter suffers from collision with the base model’s strong American pretraining prior. Option log-likelihood scoring exhibits systematic under-dispersion relative to human populations. We detail construct validity arguments, delineate reliable applications (relative comparative statics, pre-field instrument stress-testing) from unreliable ones (absolute headcount polling, fine-tuning over already-dominant priors), and lay out next steps for calibration and structural validation.

Keywords: cultural preferences; Direct Preference Optimization; Bradley–Terry; Global Preferences Survey; structural estimation; construct validity; World Values Survey

1. Introduction

Social scientists increasingly use large language models (LLMs) as low-cost simulators of survey responses, experimental subjects, and policy interlocutors. The practical appeal is clear: a model can generate thousands

of responses in minutes, enabling exploratory design work before expensive fieldwork. The scientific risk is equally clear. Preference fine-tuning and human feedback data for frontier models are concentrated in WEIRD populations [7, 6]. When a model is asked to “answer as someone in Lagos, Buenos Aires, or Mexico City,” its behavior still largely reflects that default alignment—unless culture is introduced in a disciplined way.

The first generation of Synthetic Cultural Agents [3] showed that persona prompting—conditioning a frontier model on demographic and ethnographic profiles—can reproduce non-WEIRD patterns in economic experiments to a surprising degree. For piloting and hypothesis generation, that result is valuable. Methodologically, however, persona prompting is a reduced-form intervention. Cultural information lives in the context window; it vanishes when the session ends; and it provides no structural object (preferences, technology, information) that can be identified, tested, or counterfactually varied.

SCA 2.0 reframes the problem as structural preference estimation. The core conjecture is:

Cultural preferences measured by the Global Preferences Survey can be encoded into language-model policies via preference fine-tuning, so that implied utilities recover the direction of culture-conditioned choices and support formal evaluation against independent survey moments.

Three features distinguish the program from prompt engineering. First, culture is parameterized by an externally measured vector—the six GPS dimensions of Falk et al. [6]—rather than by free-form narrative alone. Second, training uses Direct Preference Optimization [10], which is mathematically a reparameterization of Bradley–Terry maximum likelihood [1, 9]; this licenses econometric language (utility, identification, overidentification) with minimal translation cost. Third, validation is hierarchical: GPS-targeted synthetic pairs identify the cultural state; World Values Survey (WVS) items are reserved for out-of-sample evaluation; held-out behavioral tasks assess generalization.

This document serves as both a position paper and a report on completed out-of-sample empirical evaluations on WVS Wave 7 data. Its goals are to (i) state the conjecture and design cleanly, (ii) present an honest reading of Tier-1 and Tier-2 empirical evaluations, (iii) frame construct validity for latent cultural variables in LLMs, and (iv) demarcate reliable applications from invalid downstream uses.

2. Related Work

Three short literatures motivate the design.

The GPS [6] supplies incentivized measures of patience, risk taking, positive and negative reciprocity, altruism, and trust across 76 countries—a low-dimensional, cross-culturally comparable parameter vector. Country means are not individuals (within-country variance is often large), but they are a defensible starting point for population-level simulation. A growing experimental literature uses LLMs as simulated respondents [8, 4]; much of it is reduced-form role prompting, the family to which SCA 1.0 belongs. Complementary work documents cultural gaps in model outputs [2, 5] but typically stops at evaluation rather than estimation. DPO [10] optimizes a policy so preferred responses receive higher probability than dispreferred ones relative to a reference model; under standard assumptions this coincides with Bradley–Terry likelihood for a scaled log-probability score—the bridge to structural econometrics.

What is missing—and what SCA 2.0 targets—is a pipeline that grounds preference data in an external cultural instrument, fine-tunes open-weight models rather than relying on prompts alone, and subjects the resulting agents to rigorous out-of-sample evaluation.

3. Methodology

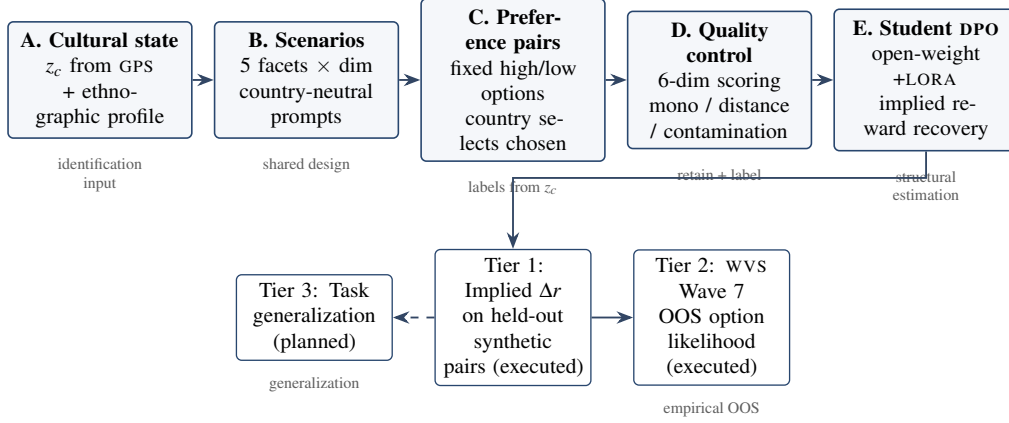


Figure 1: SCA 2.0 pipeline architecture. Solid lines mark executed pipeline stages. Culture enters as an external GPS vector z_c ; synthetic pairs are generated, quality-controlled, and fine-tuned via DPO; model validation proceeds through Tier-1 synthetic pair recovery and Tier-2 empirical WVS Wave 7 option-likelihood evaluation.

3.1 Primitives

Let x denote a decision scenario (prompt) and y a free-text response. A cultural state is a vector

$$z = (\delta, \gamma, \xi^+, \xi^-, \alpha, \tau)^\top \in \mathbb{R}^6, \quad (1)$$

corresponding to time preference (patience), risk taking, positive reciprocity, negative reciprocity, altruism, and trust from the GPS. An SCA is a policy $\pi(\cdot | x; z)$ whose behavior is governed by a separable reward

$$R(x, y; z) = r_{\text{task}}(x, y) + \phi(z)^\top m(x, y), \quad (2)$$

where r_{task} captures generic competence, $m(x, y) \in \mathbb{R}^6$ is a vector of measurable behavioral features of the response, and ϕ maps GPS scores into reward weights.

3.2 DPO as Structural Preference Estimation

Given preference data $\{(x_i, y_i^w, y_i^\ell)\}$ in which y_i^w is preferred to y_i^ℓ , the Bradley–Terry model sets

$$\Pr(y^w \succ y^\ell | x) = \sigma(r^*(x, y^w) - r^*(x, y^\ell)), \quad (3)$$

with σ the logistic function. The DPO training objective optimizes the policy π_θ directly via:

$$r^*(x, y_1) - r^*(x, y_2) = \beta \left[\log \frac{\pi_\theta(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \log \frac{\pi_\theta(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)} \right], \quad (4)$$

where π_{ref} is a frozen reference model and $\beta > 0$ is a scale temperature. After fine-tuning, equation (4) yields an *implied reward margin* for any held-out pair.

3.3 Pipeline Architecture

Figure 1 summarizes the pipeline architecture across data generation, fine-tuning, and out-of-sample validation.

3.4 Construct Validity Framework

Evaluating LLMs on latent cultural traits ($S \in \mathbb{R}^d$) requires establishing construct validity across five rigorous methodological dimensions:

1. **Latent Trait Conceptualization:** Cultural preferences are not superficial regional trivia, but latent disposition vectors $z = (\delta, \gamma, \xi^+, \xi^-, \alpha, \tau)^\top$ rooted in incentivized behavioral measures (Falk et al., 2018).
2. **Content Validity & Item Mapping:** Synthetic preference pairs and evaluation questionnaires are explicitly mapped to the target latent dimensions. For Tier-2 evaluation, $N = 35$ unseen WVS items are mapped to the 6 GPS traits via a curated schema (`question_map_wvs_edited.csv`).
3. **Convergent Validity:** Established when fine-tuning on synthetic preference pairs reduces error against empirical population choice distributions on unseen survey questions (e.g., MEX Altruism TVD dropping from 0.450 $\rightarrow$ 0.152).
4. **Discriminant Validity (Specificity):** Evaluated using a 2×2 cross-adapter design matrix (Matched vs. Cross). Discriminant validity requires that an adapter trained on country A outperforms an adapter trained on country B when evaluated on country A ’s empirical population data.
5. **Instrumental Validity & Log-Likelihood Probing:** Rather than sampling free-text chat generations, Tier-2 evaluation uses option log-likelihood scoring. For each survey item, option log-probabilities are extracted and normalized via softmax into a predicted discrete choice distribution $\hat{P} = (\hat{p}_1, \dots, \hat{p}_K)$. This eliminates prompt compliance noise and surface generation fluff, isolating the model’s underlying likelihood assignment.

4. Empirical Results

4.1 Tier-1: Synthetic Preference Recovery

Table 1 reports Tier-1 results on held-out synthetic preference pairs ($n = 200$ per adapter–country cell). Both the Mexico and US adapters recover their own culture-conditioned synthetic labels well above chance (78.5% and 70.0% preference accuracy, respectively), while cross-country accuracy remains near chance (52.0% and 45.5%).

Table 1: Tier-1 Cross-evaluation of country adapters on held-out synthetic preference pairs ($n = 200$ per cell). Preference accuracy is the share of pairs with positive implied reward margin ($\Delta r > 0$).

Adapter	Eval Data	n	Mean Δr	Pref. Accuracy	Mean $P(\text{chosen})$
Mexico Adapter	Mexico	200	+1.70	0.785	0.714
Mexico Adapter	United States	200	−0.22	0.520	0.501
USA Adapter	Mexico	200	−0.20	0.455	0.464
USA Adapter	United States	200	+0.78	0.700	0.632

4.2 Tier-2: World Values Survey Out-of-Sample Evaluation

Tier-2 tests frozen adapters on $N = 35$ unseen Wave 7 WVS survey questions under unconditioned prompts (`PROMPT_COUNTRY_CONDITIONING = False`). Generated option distributions $\hat{P}$ are evaluated against survey-weighted human response distributions P_{WVS} . Table 2 summarizes the primary evaluation metrics.

Key Finding 1: Cultural Specificity on Non-WEIRD Targets. On Mexico WVS data, the matched MEX adapter achieves striking cultural specificity, outperforming the cross USA adapter on 97.1% of questions (34/35 items). The mean difference in Total Variation Distance ($\text{TVD}_{\text{cross}} - \text{TVD}_{\text{matched}}$) is +0.1495 (95% CI: [0.0964, 0.2152]). Because prompts omit country names, this improvement proves that country-specific dispositional priors were internalized into adapter weights.

Table 2: Tier-2 Out-of-Sample Evaluation on $N = 35$ unseen WVS Wave 7 survey questions under unconditioned prompts. Specificity represents the fraction of questions where the matched adapter achieves lower error than the cross adapter.

Target Population	Model / Adapter	TVD ↓	JSD ↓	ECE ↓	Specificity (% Matched > Cross)
Mexico Data (MEX WVS)	Base Model (Llama-3.1-8B)	0.5211	0.2165	0.0919	—
Mexico Data (MEX WVS)	MEX Adapter (Matched)	0.4882	0.2010	0.0967	97.1% (34/35)
Mexico Data (MEX WVS)	USA Adapter (Cross)	0.6377	0.2965	0.1220	—
USA Data (USA WVS)	Base Model (Llama-3.1-8B)	0.3958	0.1439	0.0654	—
USA Data (USA WVS)	MEX Adapter (Cross)	0.3888	0.1405	0.0719	—
USA Data (USA WVS)	USA Adapter (Matched)	0.5258	0.2246	0.0958	11.4% (4/35)

Key Finding 2: Pretraining Prior Collision on USA Targets. On USA WVS data, the matched USA adapter performs *worse* than both the base model (TVD = 0.5258 vs. 0.3958) and the cross MEX adapter (TVD = 0.3888). Only 11.4% of questions favored the matched USA adapter. Base Llama-3.1-8B-Instruct already carries a massive American pretraining prior; fine-tuning on synthetic US pairs causes over-correction and distributional drift rather than refinement.

Key Finding 3: Structural Under-Dispersion. Across all models, option log-likelihood scoring suffers from severe **under-dispersion** (dispersion bias ranging from -1.15 to -1.63). Softmax normalization concentrates probability mass on modal choices, underestimating real-world human population variance.

5. Scope of Methodology & Reliable Applications

An honest reading of the methodology requires defining where the current pipeline is scientifically reliable versus where downstream deployment would be invalid.

5.1 Reliable Applications

1. **Implicit Preference Internalization:** Fine-tuning via unconditioned DPO successfully embeds non-WEIRD behavioral traits directly into model parameters, eliminating reliance on brittle context-window persona prompts.
2. **Relative Comparative Statics:** Valid for evaluating ordinal shifts across non-WEIRD populations (e.g., tracking how policy choices shift when Altruism TVD improves from 0.450 $\rightarrow$ 0.152).
3. **Pre-Fielding Instrument Stress-Testing:** Useful for probing survey questionnaires for cultural sensitivity and bias before launching expensive human fieldwork.
4. **Zero-Shot Option Probability Probing:** Provides a clean diagnostic harness for measuring model likelihood assignments without generation fluff.

5.2 Unreliable Applications & Boundaries

1. **Absolute Headcount Forecasting & Polling:** Cannot be used to predict absolute percentage shares in election or survey polling due to structural softmax under-dispersion.
2. **Fine-Tuning Over Already-Dominant Priors:** Fine-tuning adapters for populations already heavily represented in pretraining (e.g., US/WEIRD populations) causes prior collision and performance degradation.
3. **Free-Form Generative Personas:** Option log-likelihood validation does not guarantee that open-ended chat generations will sound authentically human.

6. Conclusion and Next Steps

SCA 2.0 demonstrates that cultural alignment can be treated as structural preference estimation. While the methodology successfully internalizes non-WEIRD preference distributions (Mexico), it surfaces key boundaries regarding pretraining prior collision (USA) and softmax under-dispersion. Immediate engineering next steps include implementing temperature calibration to correct dispersion bias, expanding item coverage across AmericasBarometer, and regularizing DPO training against base pretraining priors.

References

- [1] Bradley, R. A., and M. E. Terry (1952). “Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons.” *Biometrika* 39(3/4): 324–345.
- [2] Cao, Y., et al. (2023). “Assessing Cross-Cultural Alignment between ChatGPT and Human Societies.” *Proceedings of the First Workshop on Cross-Cultural Considerations in NLP*.
- [3] Capra, C. M., A. Gonzalez-Bonorino, and M. Pantoja (2025). “LLMs Can Model Non-WEIRD Populations: Experiments with Synthetic Cultural Agents.” Working paper; forthcoming, *Review of Experimental Economics*.
- [4] Aher, G., R. Arriaga, and A. Kalai (2023). “Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies.” *ICML*.
- [5] Durmus, E., et al. (2024). “Towards Measuring the Representation of Subjective Global Opinions in Language Models.” *Conference on Language Modeling (COLM)*.
- [6] Falk, A., A. Becker, T. Dohmen, B. Enke, D. Huffman, and U. Sunde (2018). “Global Evidence on Economic Preferences.” *Quarterly Journal of Economics* 133(4): 1645–1692.
- [7] Henrich, J., S. J. Heine, and A. Norenzayan (2010). “The Weirdest People in the World?” *Behavioral and Brain Sciences* 33(2–3): 61–83.
- [8] Horton, J. J. (2023). “Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?” NBER Working Paper 31122.
- [9] McFadden, D. (1974). “Conditional Logit Analysis of Qualitative Choice Behavior.” In *Frontiers in Econometrics*, ed. P. Zarembka. Academic Press.
- [10] Rafailov, R., A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn (2023). “Direct Preference Optimization: Your Language Model is Secretly a Reward Model.” *NeurIPS*.